Results and Discussion

Descriptive Statistics

The analysis was conducted using six major stocks traded on Borsa Istanbul: ASELSAN (ASELS), Turkish Airlines (THYAO), BIM Stores (BIMAS), Garanti Bank (GARAN), Tupras (TUPRS), and Koc Holding (KCHOL). To represent the overall Turkish stock market, the BIST100 index was used as the benchmark.

The descriptive statistics revealed noticeable differences in both return and risk characteristics among the selected stocks. While some stocks generated higher average daily returns, they were also associated with higher volatility. This finding highlights the fundamental risk-return tradeoff observed in financial markets.

Among the selected companies, THYAO and TUPRS generally exhibited stronger growth characteristics during the examined period. Their higher average returns indicate that investors were rewarded for taking additional risk. However, these stocks also displayed larger fluctuations in daily returns compared to more defensive investments.

BIMAS showed relatively stable behavior throughout the period. As a consumer staples company, its returns tended to fluctuate less during periods of market uncertainty. This lower volatility makes it comparatively more defensive than cyclical stocks such as THYAO or TUPRS.

GARAN and KCHOL demonstrated performance patterns closely linked to overall economic conditions. Since banking and holding companies are strongly affected by interest rates, inflation expectations, and macroeconomic developments, their return distributions reflected broader market sentiment.

ASELS displayed periods of strong upward momentum, reflecting the increasing importance of the defense industry in Türkiye. Although the stock experienced substantial growth during certain periods, it also showed occasional spikes in volatility.

Normalized Price Performance Analysis

Figure 1 presents the normalized stock prices, where all assets begin at a value of 100. This visualization allows direct comparison of long-term performance regardless of differences in original stock prices.

The figure shows that not all stocks followed the same growth trajectory. THYAO and TUPRS generally achieved stronger long-term appreciation than the BIST100 benchmark, indicating that these companies outperformed the broader Turkish market during the selected period.

ASELS also demonstrated significant growth potential, particularly during periods when defense-related investments gained increased attention. However, the stock occasionally exhibited sharp upward and downward movements, reflecting changing investor expectations.

BIMAS maintained a smoother growth trend relative to most other stocks. This behavior is consistent with its position in the retail sector, where demand tends to remain relatively stable across different economic environments.

GARAN and KCHOL followed trajectories that were more closely aligned with overall market movements. Their performance generally reflected broader macroeconomic conditions rather than company-specific growth shocks.

Overall, the normalized price chart indicates that several selected stocks were able to outperform the BIST100 benchmark over the long run, suggesting that stock selection may provide additional returns beyond passive market exposure.

Cumulative Return Analysis

The cumulative return chart provides insight into the wealth accumulation effect of holding each stock over time.

The results indicate substantial differences in long-term investment outcomes. THYAO generated one of the strongest cumulative return profiles, reflecting the recovery of travel demand and operational improvements in the airline industry. TUPRS also achieved significant cumulative gains, benefiting from developments in energy markets and refining operations.

ASELS showed strong cumulative growth during periods characterized by increased defense spending and positive sector expectations. Nevertheless, the stock experienced intermittent periods of correction, illustrating the importance of considering both return and volatility.

BIMAS delivered more moderate but stable cumulative returns. While it did not always produce the highest gains, its consistency may appeal to investors seeking lower-risk opportunities.

GARAN and KCHOL produced cumulative returns that generally moved with the overall market. Their performance reflects the close relationship between financial institutions, conglomerates, and macroeconomic developments within Türkiye.

The comparison with BIST100 demonstrates that several selected stocks exceeded market performance during the sample period.

Return Distribution Analysis

Histograms of daily returns reveal that stock returns are not perfectly normally distributed. Most observations are concentrated around zero, while extreme positive and negative returns occur less frequently.

THYAO and TUPRS exhibited wider distributions, indicating greater volatility and a higher probability of large daily price movements. Such behavior is common among cyclical industries that are highly sensitive to economic conditions.

BIMAS displayed a narrower return distribution, suggesting lower volatility and more stable daily performance. This result supports the defensive characteristics typically associated with retail companies.

ASELS occasionally generated large return observations in both directions, producing a wider tail structure in its distribution. This suggests higher uncertainty but also greater potential upside opportunities.

GARAN and KCHOL demonstrated return distributions that were relatively balanced but still exhibited occasional extreme observations, reflecting market-wide shocks and economic events.

Box Plot Analysis

The box plots provide additional evidence regarding the variability of returns.

THYAO and TUPRS generally showed larger interquartile ranges and more outliers than the other stocks, confirming their higher volatility. Investors holding these assets may experience larger gains but also greater losses.

BIMAS displayed a more compact return distribution with fewer extreme values. This finding is consistent with the stock's relatively stable business model.

ASELS showed several extreme observations, indicating periods of unusually high market activity. Such outliers may be associated with significant news events, government announcements, or changes in investor expectations.

GARAN and KCHOL occupied a middle position between defensive and aggressive investments, exhibiting moderate variability and occasional outliers.

Correlation Analysis

The correlation matrix indicates that most selected stocks share positive relationships with one another. This is expected because all companies operate within the same national economy and are influenced by common macroeconomic factors.

The strongest correlations were generally observed between GARAN and KCHOL, as well as between GARAN and the BIST100 benchmark. Banking and holding companies often react similarly to interest rate decisions, inflation expectations, and overall economic growth.

BIMAS exhibited comparatively lower correlations with some cyclical sectors, suggesting potential diversification benefits when included in a portfolio.

Although diversification opportunities exist, the correlation results indicate that market-wide events continue to affect all selected assets to some degree.

Hypothesis Testing Results

To evaluate whether the selected stocks produced returns significantly different from the overall market, paired t-tests were conducted using daily returns and the BIST100 benchmark.

The null hypothesis stated that the average daily return of a stock is equal to the average daily return of BIST100. The alternative hypothesis stated that a significant difference exists between the two returns.

A paired t-test was chosen because stock returns and benchmark returns are observed on the same trading days and therefore represent dependent observations.

The results show that some stocks exhibit statistically significant differences relative to the benchmark, while others move more closely with overall market performance. Stocks with significant results suggest the presence of company-specific performance drivers beyond general market conditions.

These findings demonstrate that stock selection can meaningfully influence investment performance and that not all stocks simply replicate the behavior of the broader market.

Overall Findings

The EDA reveals substantial variation in performance, risk, and return characteristics among the selected stocks. THYAO and TUPRS generally provided stronger growth potential but were associated with higher volatility. BIMAS offered more stable returns with lower risk. ASELS demonstrated strong growth opportunities combined with periods of elevated uncertainty. GARAN and KCHOL closely reflected broader economic and market conditions.

The use of BIST100 as the benchmark provided a realistic representation of the Turkish equity market and enabled more meaningful comparisons than using the average return of selected stocks. Overall, the analysis suggests that careful stock selection may lead to performance outcomes that differ significantly from passive market investment strategies.